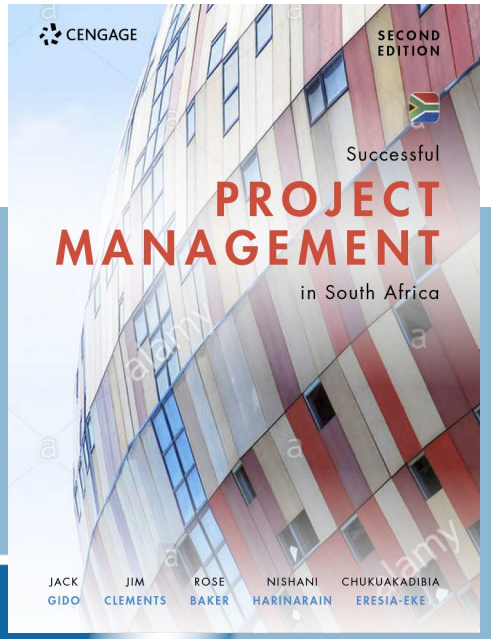
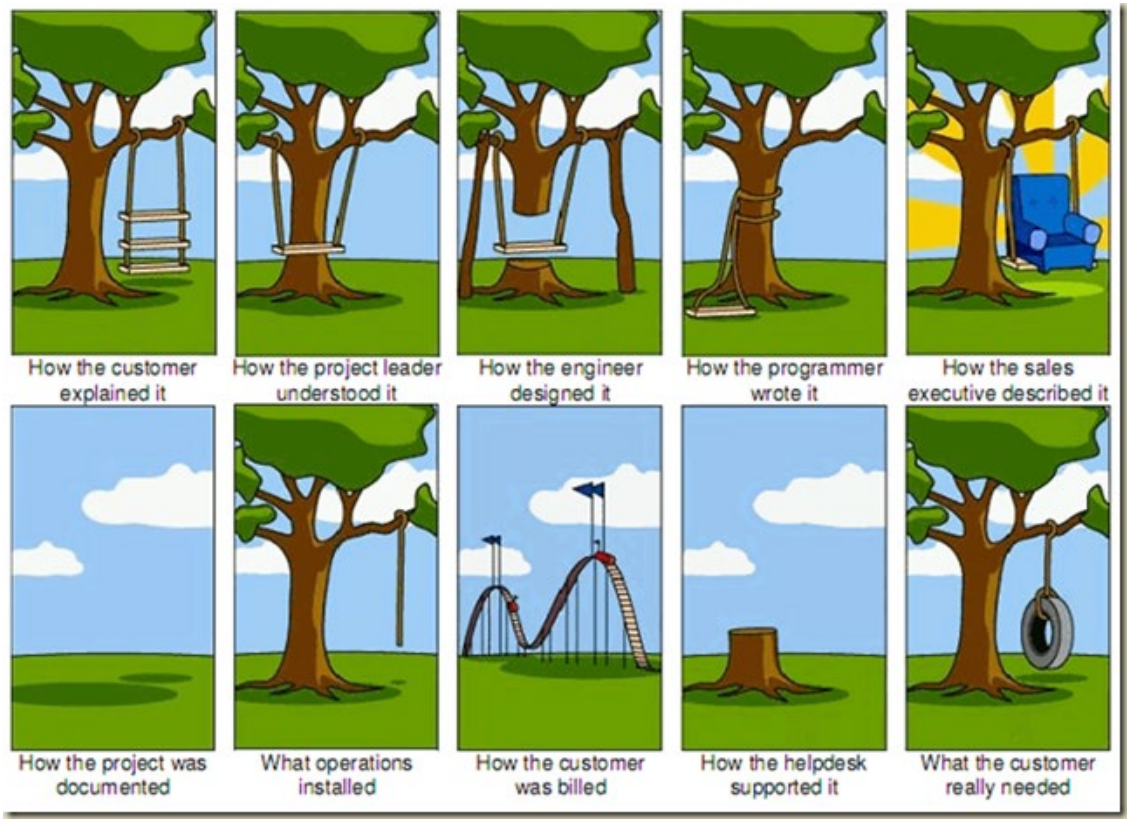
CHAPTER 4



Defining Scope, Quality, Responsibility and Activity Sequence

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In the previous learning units, you explored the **first phase** of the **project life cycle**, i.e. **initiating** a project, which includes **identifying** and **selecting** projects, preparing **RFPs** and concepts related to project **proposals** and **contracts**.

Before jumping in and starting the project, the **next step** is to **develop** a project **plan**. **Planning** determines **what** needs to be done, **how** it will get done, **who** will do it, **how long** it will take, how much it will **cost** and what the **risks** are. In this learning unit, we explore the **development of a well thought out plan**.

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Chapter Concepts

- Clearly defining the project objective
- Preparing a project scope document
- Understanding the importance of planning for quality
- Creating a work breakdown structure
- Assigning responsibility for work items
- Defining specific activities
- Creating a network diagram
- Utilising a project management methodology called the *systems development life cycle for information systems development projects*

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Learning Outcomes

- Establish a clear project objective
- Prepare a project scope document
- Discuss the importance and elements of a project quality plan
- Develop a work breakdown structure
- Prepare a responsibility assignment matrix
- Describe how to define specific activities
- Create a network diagram

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Performance Domains of PMBOK® 7th edition covered in this chapter:	Knowledge Areas of PMBOK® 6th edition covered in this chapter:
• Team • Planning • Project Work • Delivery • Measurement	• Project Integration Management • Project Scope Management • Project Quality Management • Project Time Management

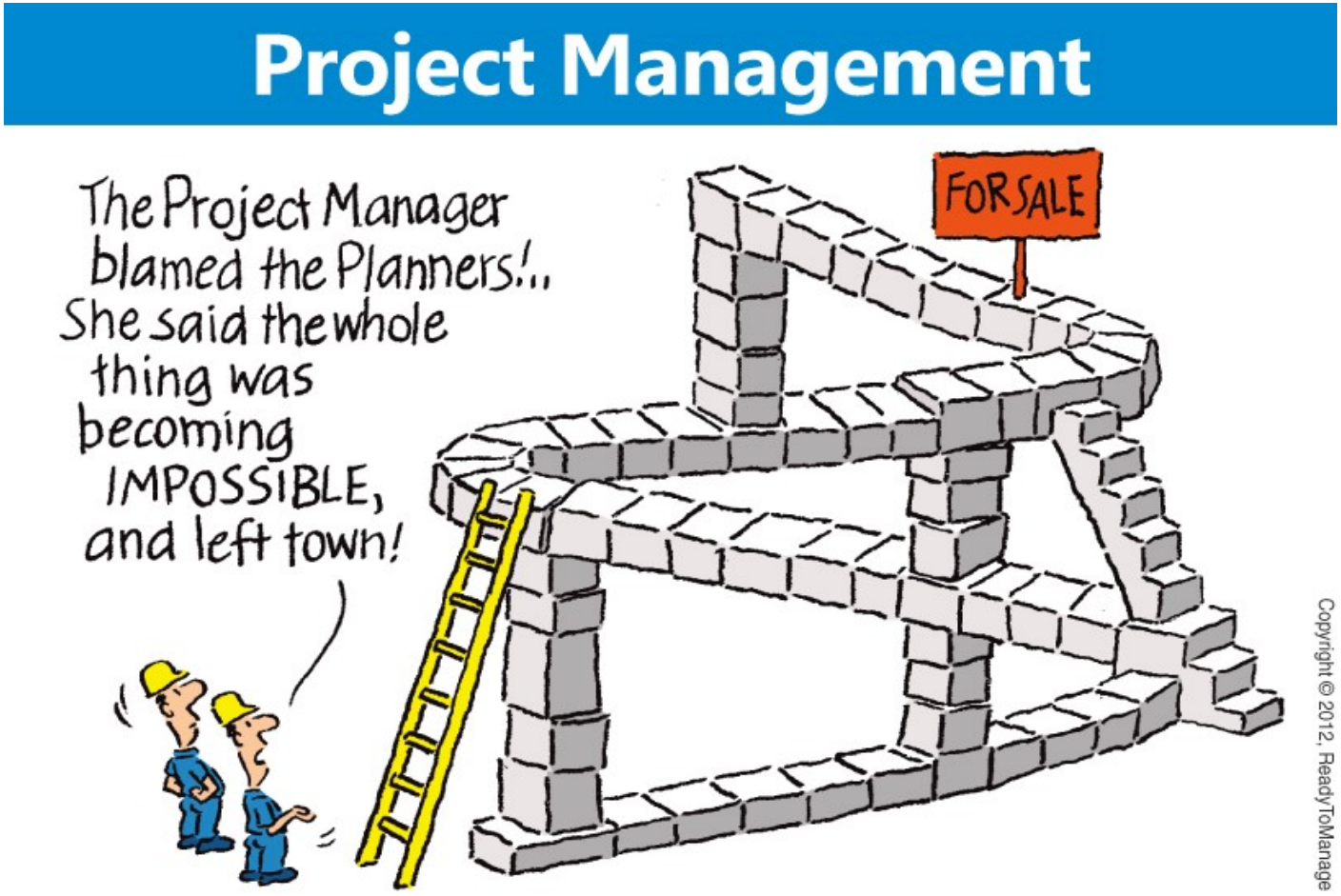
Project Management Principles relevant to this chapter

Principle 7 (Tailoring Based on Context)
Principle 8 (Quality)

Principle 8 – Quality: Ensuring quality on a project means meeting and or exceeding the stakeholder's expectations and delivering a project that conforms to the project objectives by focusing on preventing and errors and defects.

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- Have you ever **started** a **task** and then felt it was bigger than you thought, and the **deadline** is **creeping up** on you?
  - How do you **cope** with it then?
  - What could you do in **future** to **avoid** such anxiety?
- Why do you think **project managers** may become anxious when **small adjustments or changes keep on creeping** up in a project?



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Establish Project Objective

- Planning process is based on the project objective
  - Establishes what is to be achieved
  - Often stated in the project charter or RFP
  - Is the tangible end product
- Project objective includes
  - Expected benefits
  - Primary project end product or deliverable
  - Date required to be completed
  - Budget
- Changes agreed upon by customer and contractor
- Examples of Project Objectives on page 108

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Define Project Scope

Project scope

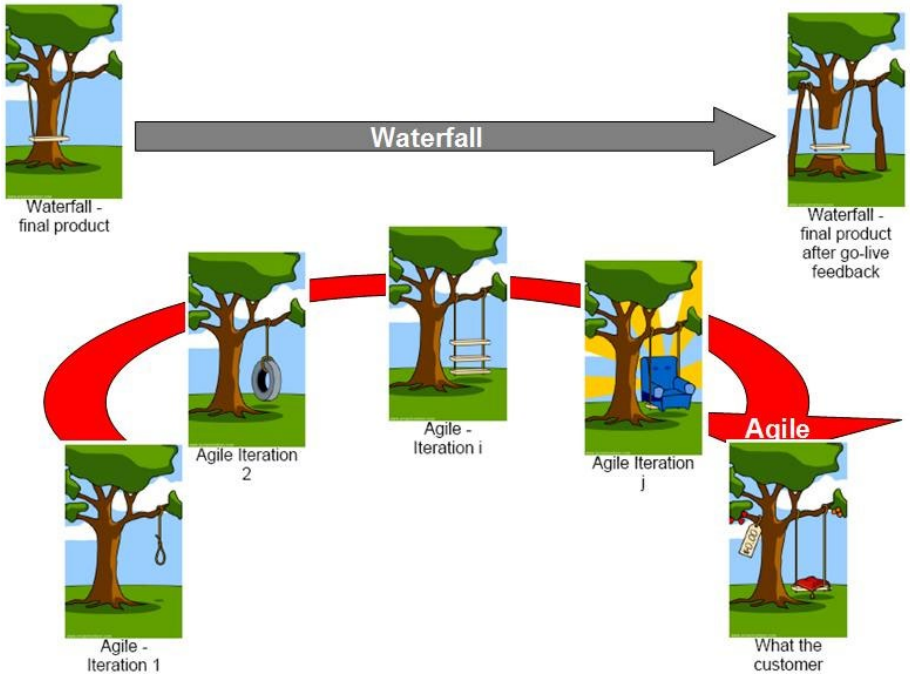
- Defines what needs to be done
- Includes
  - Items contained in project charter, RFP, proposal
  - More detail
- Establishes common understanding of scope with stakeholders

Project scope document

- Usually contains
  - Customer requirements
  - Statement of Work (*SOW*)
  - Deliverables
  - Acceptance criteria
  - Work Breakdown Structure (*WBS*)
- Establishes baseline
- Change control system to avoid scope creep

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Self-Research: Agile Project Scope VS Traditional Project Scope



- What is **agile project scope**?
  - How is it **different** from a **traditional** project scope in project management?
- How is agile project scope management, **controlled**?
- How is **scope creep managed** in agile projects?

http://itsadeliverything.com/agile-project-scope

http://itsadeliverything.com/controlled-scope-management

https://agilepm.com/managing-scope-creep-in-agile-projects

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- Which **shoe brand** do you **prefer** buying? Why?
  - What do you do if you do **not** get your **preferred brand** when buying shoes?
- What does it mean to say that a project is ***worth the client's time and money***?



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Plan for Quality

- Complete work according to specifications and standards
- Plan for assuring quality as project progresses
- Include
  - References for specifications and standards
  - Written procedures for quality tools and techniques
- Key aspects
  - Monitor as the project progresses
  - Do work right the first time rather than rework

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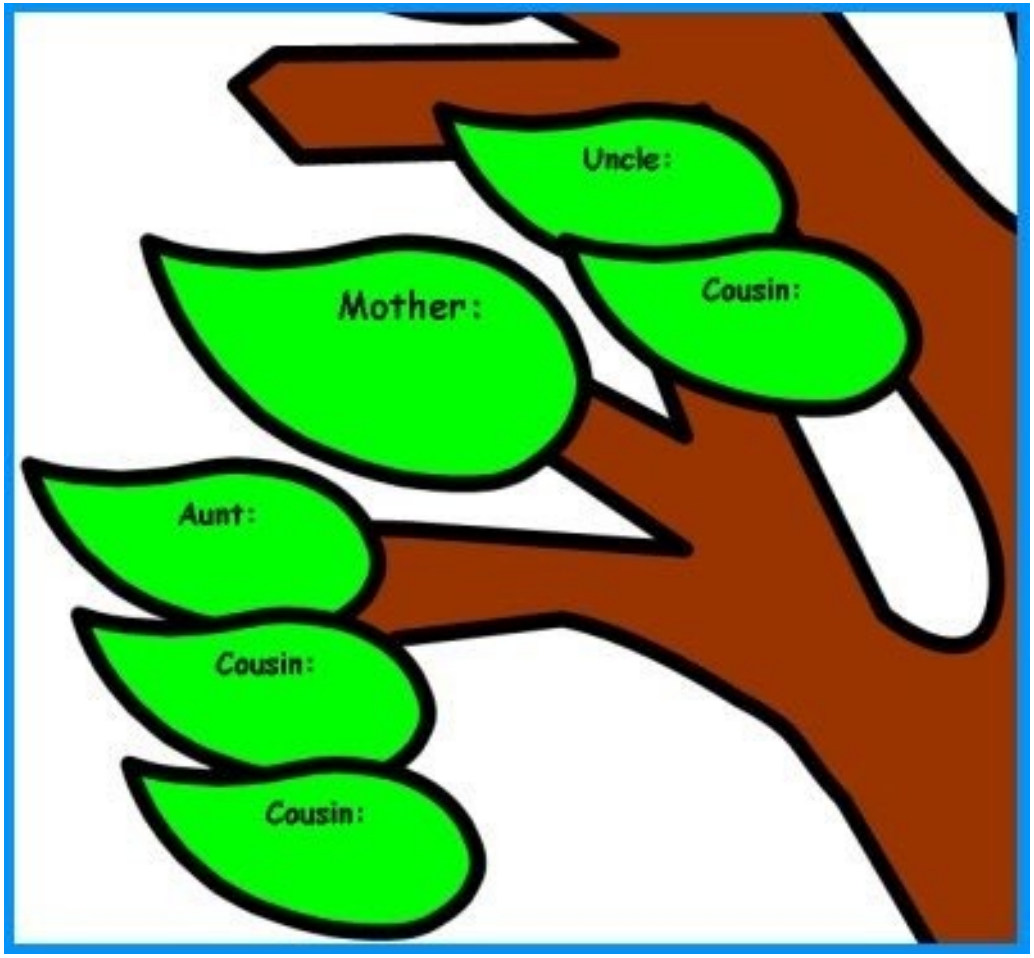
- What is a ‘family tree’?

➢ Can you map out the relationship of all the members of your family in a hierarchical structure?

➢ What information did you need about your family to ensure that this structure is correct?

- How can hierarchical structures be used to make project planning more effective?

➢ How can such hierarchical structures assist the project manager to assign responsibilities to project team members?



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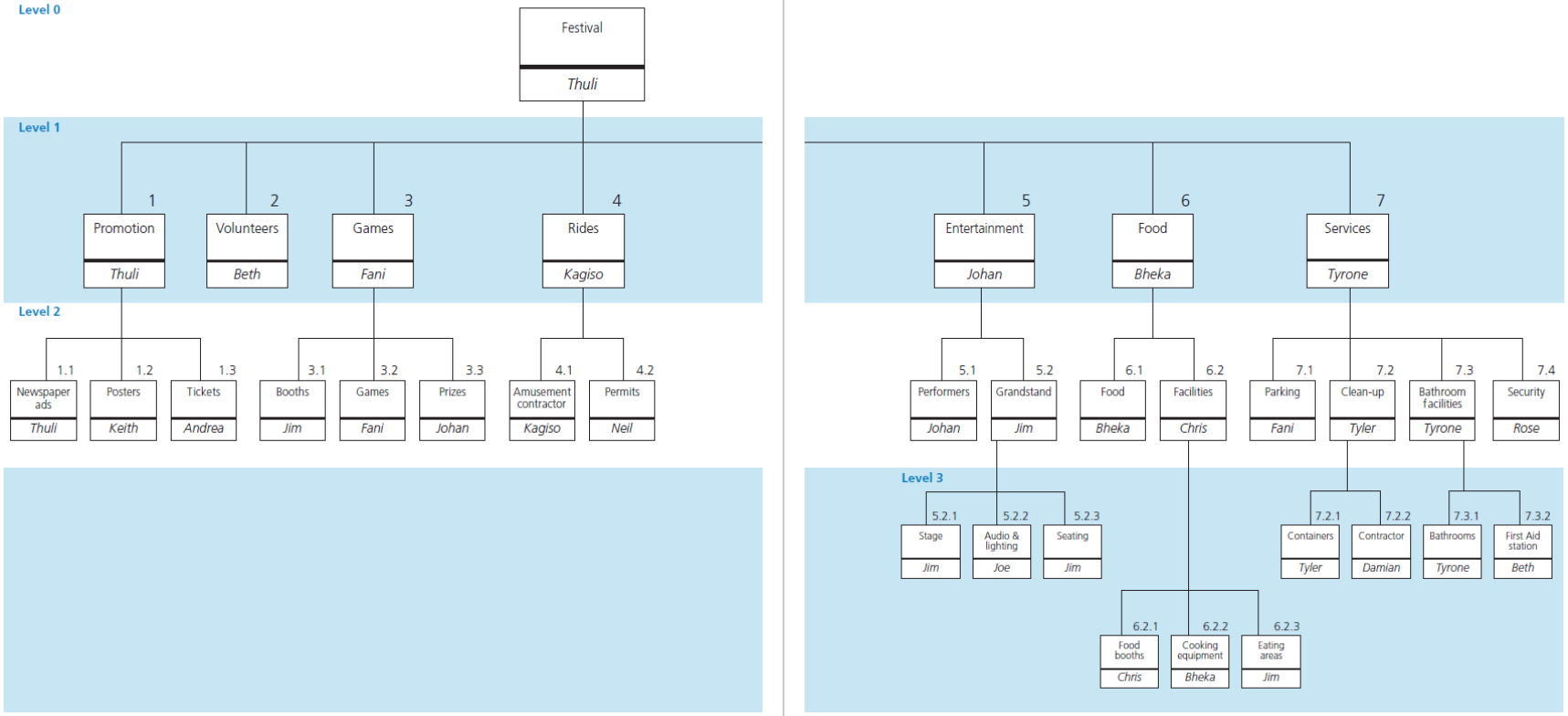
Create Work Breakdown Structure

- Deliverable-oriented hierarchical decomposition
- Organise project work and deliverables
- Create logical groupings
- Subdivide into more manageable components
  - Deliverable is output of work package
  - Resource requirements and durations can be assigned
  - Accountability can be assigned
  - Project manager can monitor and control
- Graphic chart or indentured list

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WBS with breakdown to different levels

FIGURE 4.1 Work Breakdown Structure for Festival Project

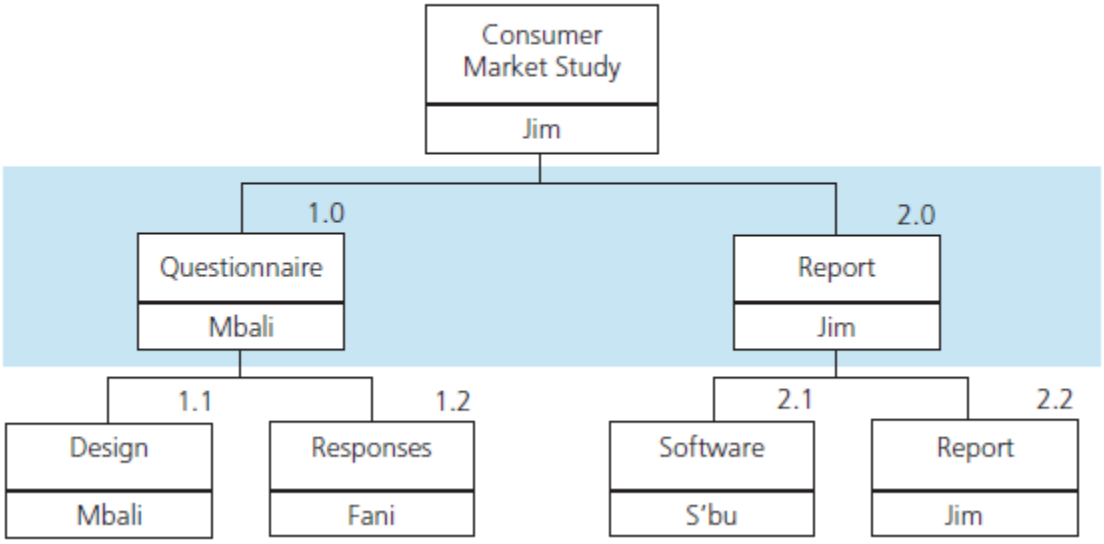


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WBS

- Graphic Chart

FIGURE 4.2 Work Breakdown Structure for Consumer Market Study Project



- Indentured List

FIGURE 4.3 Work Breakdown Indentured List for Consumer Market Study Project

WBS #	DESCRIPTION	RESPONSIBLE	DELIVERABLES
	Consumer Market Study	Jim	
1.0	Questionnaire	Mbali	
1.1	Design	Mbali	Questionnaire approved
1.2	Responses	Fani	All completed responses received
2.0	Report	Jim	
2.1	Software	S'bu	Application software working
2.2	Report	Jim	Final report

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Assign Responsibility

- Responsibility assignment matrix
  - Designate responsible individuals
    - P = Primary responsibility
    - S = Support responsibility
  - Associates responsibility
    - For each work item
    - For each individual
- Only one primary per work item

FIGURE 4.1 Work Breakdown Structure for Festival Project

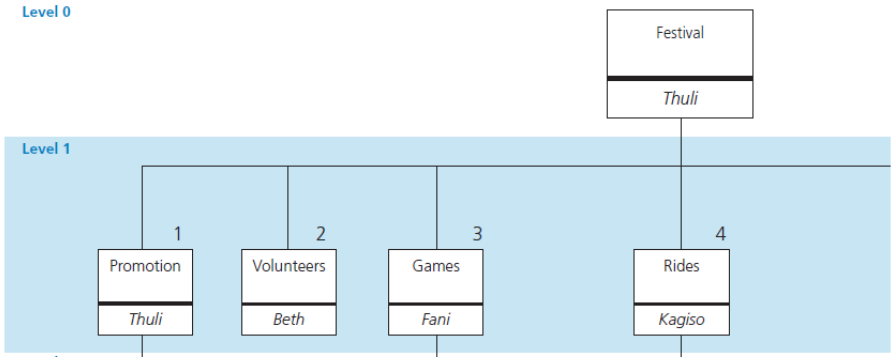


FIGURE 4.4 Responsibility Assignment Matrix for Festival Project

WBS Item	Work Item	Andrea	Beth	Bheka	Chris	Damian	Tyrone	Johan	Jim	Joe	Keith	Thuli	Neil	Kagiso	Rose	Fani	Tyler
	Festival		S	S			S	S				**P**		S		S	
1	**Promotion**	S									S	**P**					
1.1	Newspaper Ads											**P**					
1.2	Posters										**P**						
1.3	Tickets	**P**	S									S					
2	**Volunteers**		**P**						S						S		
3	**Games**							S	S							**P**	
3.1	Booths				S				**P**	S							
3.2	Games														S	**P**	
3.3	Prizes							**P**							S		
4	**Rides**												S	**P**			
4.1	Amusement Contractor													**P**			
4.2	Permits												**P**	S			
5	**Entertainment**							**P**	S	S							
5.1	Performers					S		**P**									
5.2	Grandstand								**P**	S							
5.2.1	Stage								**P**	S							
5.2.2	Audio & Lighting									**P**							
5.2.3	Seating					S			**P**								
6	**Food**		**P**		S												
6.1	Food		**P**												S		
6.2	Facilities		S		**P**				S								
6.2.1	Food Booths				**P**				S	S							
6.2.2	Cooking Equipment		**P**														
6.2.3	Eating Areas								**P**						S		
7	**Services**						**P**								S	S	S
7.1	Parking															**P**	
7.2	Clean-up					S											**P**
7.2.1	Containers																**P**
7.2.2	Contractor					**P**											
7.3	Bathroom Facilities		S				**P**										
7.3.1	Bathrooms						**P**										
7.3.2	First Aid Stations		**P**														
7.4	Security					S				S					**P**		

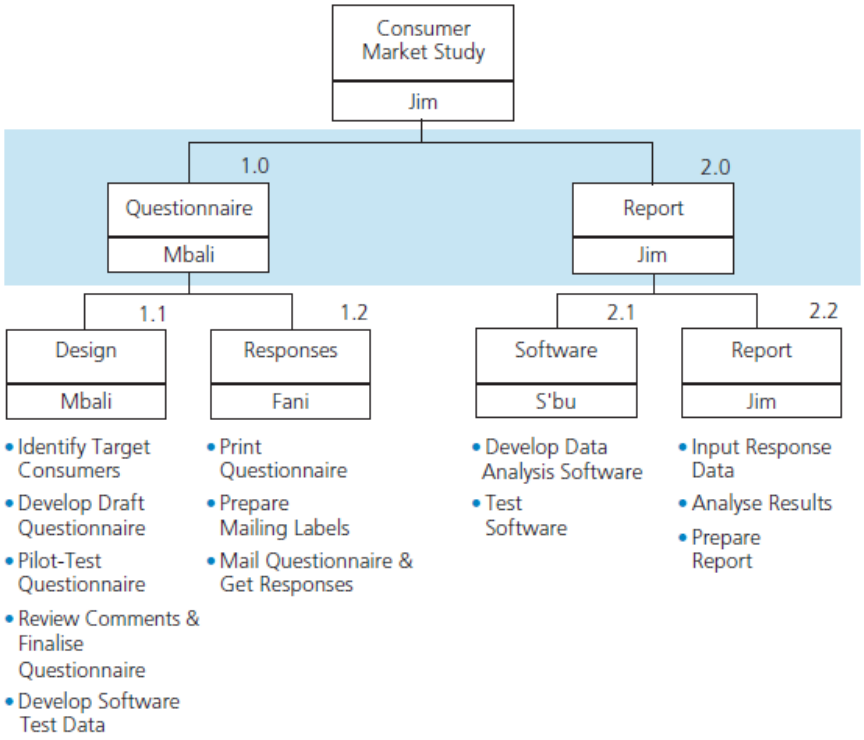
KEY: **P** = Primary responsibility; S = Support responsibility.

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Define Activities

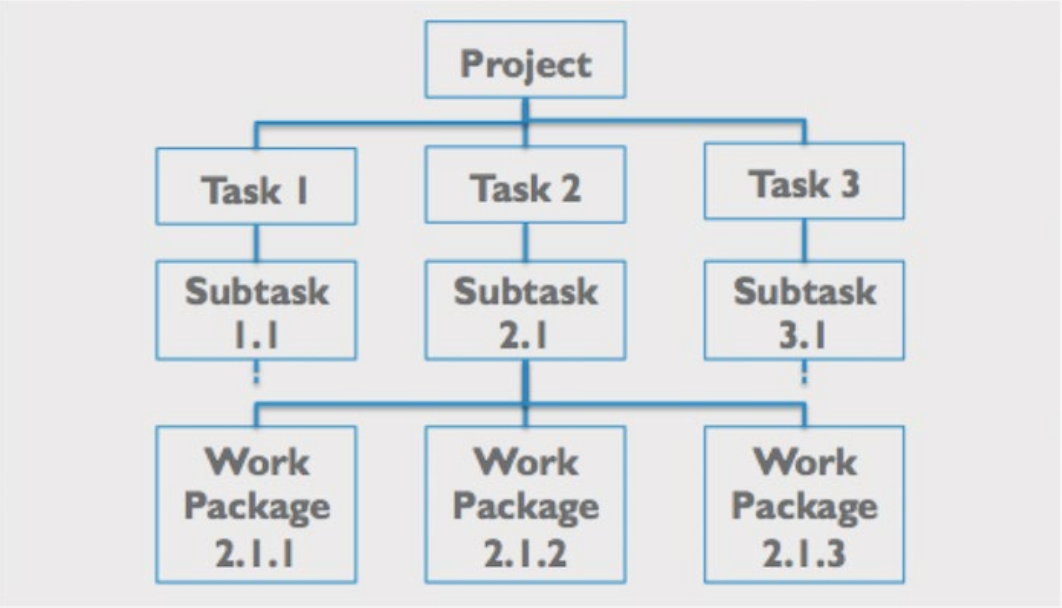
- Responsible resources define activities
  - Breakdown work packages to work items
  - Level needed to perform deliverable
  - May not be able to define all
- Comprehensive activity list
  - Not always require expenditure of effort
  - Could be wait time

FIGURE 4.5 Work Breakdown Structure for Consumer Market Study Project



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- Why do you think it is **important** that you have a **weekly time table** showing you the scheduling of modules you need to work on, the tests and the exams dates?
  - Why would it be important to **sequence** such by **dates** and **durations**?
  - What will happen if you did **not** have this time table?
- Why is it important to **sequence activities** in a **project**?



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Sequence Activities

- Network diagram
  - Defines the sequence of activities and relationships
  - Tool for arranging order
- Common techniques
  - PERT
  - CPM
  - PDM

FIGURE 4.9 Network Diagram for Consumer Market Study Project

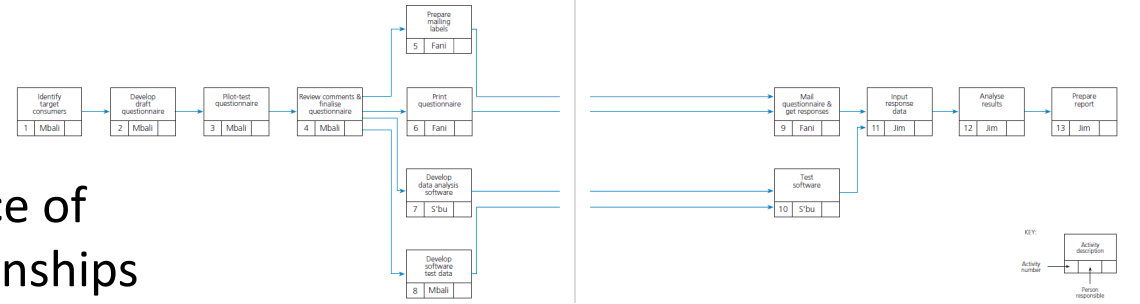
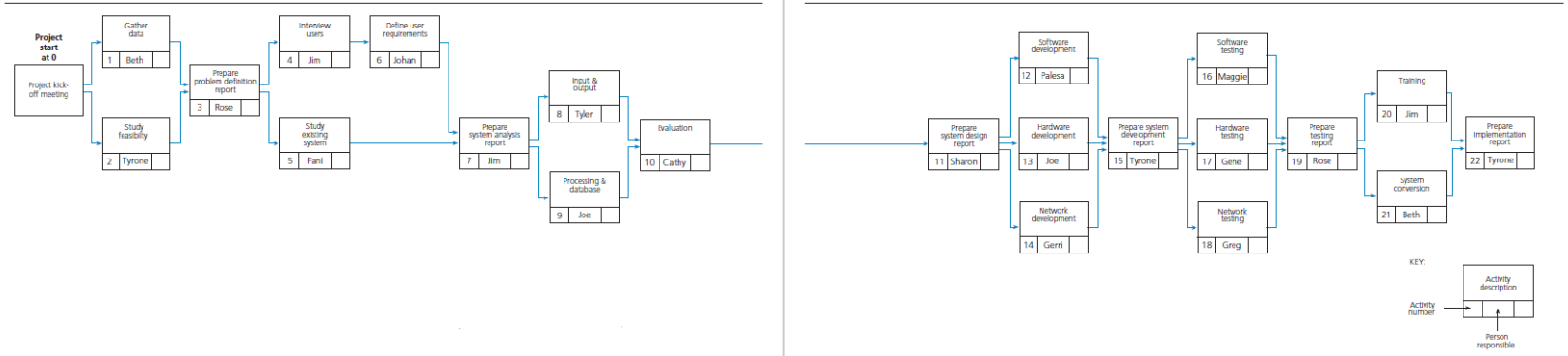


FIGURE 4.13 Network Diagram for Web-Based Reporting System Project



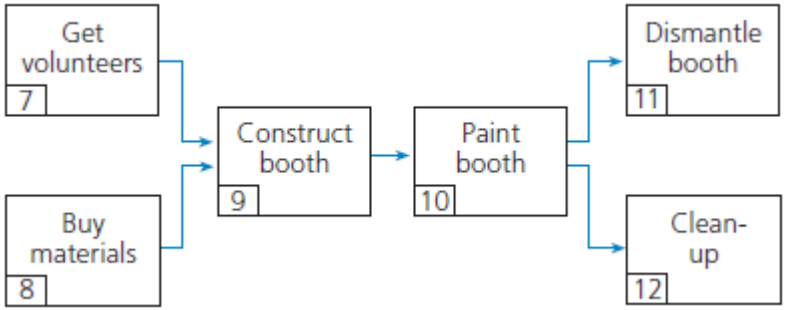
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Network principles

- Activities
  - Represented by a box
  - Consume time
  - Described by verb
- Relationships
  - Linked in serial sequence
  - Complete concurrently



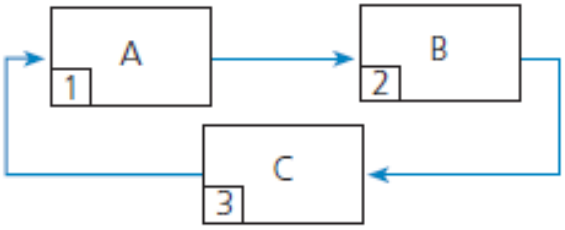




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Loops

- Illogical relationship among activities
- Perpetually repeats itself
- Not acceptable

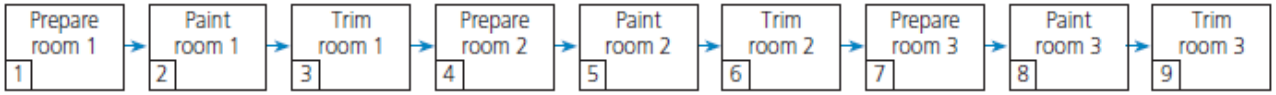


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Laddering

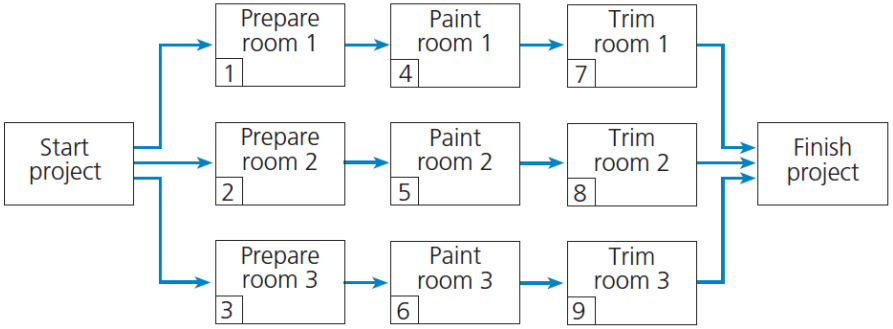
- Set of repeating activities
- Serial sequence
  - One work
  - Two wait

FIGURE 4.6 Activities Performed Serially



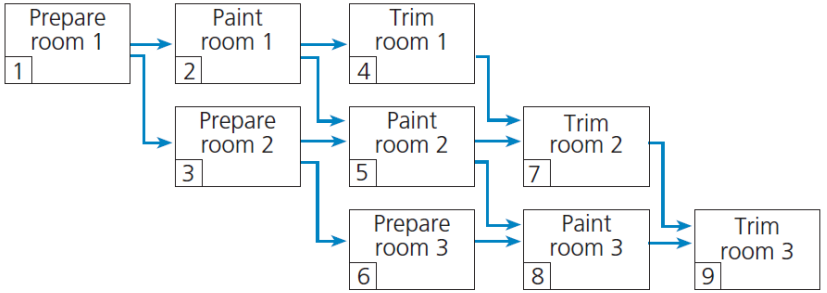
- Can be performed at same time
- Would need triple resources

FIGURE 4.7 Activities Performed Concurrently



- Ladder activities
- Allow for shortest possible time for completion
- Best use of the three workers and experts

FIGURE 4.8 Laddering



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Create network diagram

Tips and techniques

- Logical sequence of activities
- Show dependent relationships
- Elaborate as project progresses
- Use subnetworks for similar activity sequences and relationships

Development of diagram

- Questions to answer
  - Finish before next start?
  - Done concurrently?
  - Cannot be started next?
- Guidelines for detail level
  - Define each work package
  - Draw summary level network then add detail
  - Detail to responsibility or deliverable change
  - Duration less than project progress review

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Planning for Information Systems Development

Definition

- Computer-based system
  - Accepts data as input
  - Processes the data
  - Produces useful information
- Examples
  - Computerised order entry
  - E-commerce
  - Automatic teller machines
  - Billing, payroll, and inventory

Process

- Plan, execute, and control
- Systems development life cycle
  - Problem definition
  - System analysis
  - System design
  - System development
  - System testing
  - System implementation

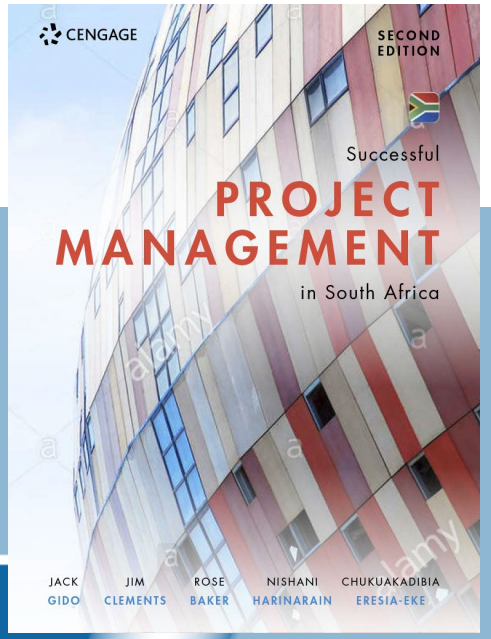
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Project Management Information Systems

- Wide variety of systems available
- Plan and control projects in interactive mode
- Plan and test different options
- Create reports, diagrams, and charts
- Interface with other software applications
- Appendix A has additional information

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CHAPTER 9



Closing the Project

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- **Have you ever had to evaluate a module at the end of a semester?**
  - **What do you think your college does with the feedback?**
- **Should projects be evaluated when they come to an end? Why/ why not?**

- **What actions should be taken during the process of How to close a project?**
  - **Why are these activities important?**
- **How will you conduct a post project evaluation?**
- **Why is it important to archive project documents?**
- **Why are some projects terminated before they are completed?**
- **What are the critical success factors relating to project planning and closing?**

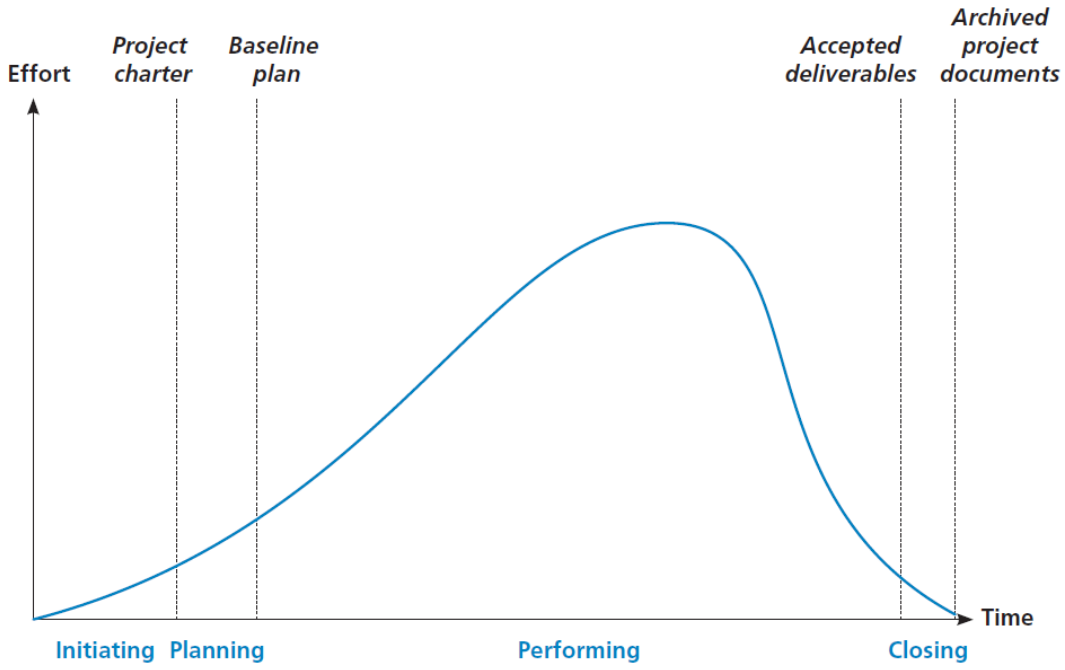


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Project Closing Actions

- Collect and make final payments
- Recognise and evaluate staff
- Conduct post project evaluation
- Document lessons learnt
- Organise and archive project documents

FIGURE 9.1 Project Life Cycle



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Final payments

- Payments from customer
  - Have activity listed to perform collection
  - Assure that all the payments have been received
- Payments to others
  - Verify all payments have been made
  - Record actual costs
- Account closing
  - Perform financial analysis on the project
  - Compare actual costs versus baseline budget

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Staff recognition and evaluation

- End project with recognition and celebration
  - Recognise team members for contributions
  - Celebrate the success of the project
- Prepare individual performance evaluations
  - Mention how knowledge expanded during project
  - Identify areas where need further development
  - Share during individual meetings or give to supervisor

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Post project evaluation

- Review and evaluate project performance
- Identify improvements
- Hold individual meetings with members
- Hold project team meeting
- Prepare written report with lessons learnt and recommendations

FIGURE 9.2 Post-project Evaluation Team Meeting Agenda

POST-PROJECT EVALUATION
Team Meeting

Agenda

1. Technical performance
   - Work scope
   - Quality
   - Managing changes
2. Budget performance
3. Schedule performance
4. Project planning and control
5. Risk management
6. Customer relationships
7. Stakeholder engagement
8. Team relationships
9. Communications
10. Problem identification and resolution
11. Lessons learnt
12. Recommendations for future projects

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Lessons learnt

- Purpose of identifying and documenting
  - Capitalise on knowledge and experience gained
  - Improve performance on future projects
- Knowledge base system or repository
  - Retrieve lessons learnt from prior projects
  - Easily access information by category or subject
- Record keeping throughout project
  - Capture lesson at time of lesson instead of end of project
  - Have as agenda topic at meetings and in training sessions

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Archive project documents

- Organise, file, and archive appropriate documents
- Organise system to make information easily retrievable
- Create database of common data elements
  - Analyse elements for trends
  - Examine records for bidding, planning, and performing future projects

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Customer Feedback

- Post project evaluation meeting with customer
  - Determine if provided the expected benefits
  - Assess the level of customer satisfaction
  - Obtain any feedback with open-ended questions
  - Discuss future opportunities
  - Secure permission to use as reference
- Participants
  - Key project team members
  - Key customer representatives
  - Project manager

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Post-project customer evaluation survey

- Analyse for trends
- Seek permission for reference

FIGURE 9.3 Post-project Customer Evaluation Survey

Post-project customer evaluation survey

Please complete this brief survey to help us evaluate and improve our project management performance. If more space is needed for responses, please attach additional pages.

Project title: _____

	Degree of satisfaction									
	Low									High
1. Completeness of work scope	1	2	3	4	5	6	7	8	9	10
Comments										
2. Quality of work	1	2	3	4	5	6	7	8	9	10
Comments										
3. Schedule performance	1	2	3	4	5	6	7	8	9	10
Comments										
4. Budget performance	1	2	3	4	5	6	7	8	9	10
Comments										
5. Communications	1	2	3	4	5	6	7	8	9	10
Comments										
6. Customer relations	1	2	3	4	5	6	7	8	9	10
Comments										
7. Overall performance	1	2	3	4	5	6	7	8	9	10
Comments										

What **benefits** did you actually realise or anticipate as a result of this project? _____

A. Quantitative benefits _____

B. Qualitative benefits _____

Suggestions on how we can improve our performance on future projects:

Name: _____ Date: _____

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Critical Success Factors

- **Plan the work and then work the plan**. It is important to develop a plan before starting to perform the project. Taking the time to develop a well-thought-out plan is critical to the successful achievement of any project.
- **Participation builds commitment**. By participating in the planning of the work, individuals will become committed to achieving it according to the plan.
- The project must have a **clear objective** of what is to be achieved. The objective should be defined in terms of end product or deliverable, schedule, and budget, and it must be agreed upon by the customer and the project team that will perform the project.
- The project scope document is valuable for establishing a **common understanding and agreement** among project stakeholders regarding the scope of the project.
- Having a quality plan at the outset of the project is extremely beneficial because it will help prevent incurring additional costs and schedule extensions due to rework caused by work and deliverables that fail to meet **quality requirements and customer expectations**.
- The key to quality control is to **monitor the quality** of the work early and regularly throughout the performance of the project, rather than waiting until all the work is completed before checking or inspecting for quality.
- The **network diagram** is also is a communication tool for the project team because it shows who is responsible for each activity and how each person's work fits into the overall project.

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Critical Success Factors

- **Recognise** the project team and celebrate achievements throughout the project.
- **Regularly ask the customer** about the level of **satisfaction** with the performance and progress of the project and take corrective action at the first hint that the customer is not satisfied.
- After the conclusion of a project, the **project performance** should be **evaluated** to learn what could be improved if a similar project were to be done in the future. Feedback should be obtained from the customer and the project team.
- **Capture and document** lessons learnt **throughout** the performance of the project.
- **Establish a knowledge base** of lessons learnt and ensure that they are communicated and used on future projects.
- **Organise** and **archive** project documentation and **analyse** key data that could be helpful in bidding, planning, or performing future projects.
- **Retain customer evaluation and feedback** information that could be valuable in developing proposals for other projects for the same customer in the future.

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Summary

- The planning process is based on the project objective, which establishes what is to be achieved.
- The project scope defines what needs to be done.
- The project scope document usually contains the customer requirements, statement of work, deliverables, acceptance criteria, and a work breakdown structure.
- The quality plan must include or reference the specifications, industry or government standards, and codes that must be used and met during the performance of the project work.
- The work breakdown structure establishes the framework for how the work will get done to produce the project deliverables.
- A responsibility assignment matrix defines who will be responsible for the work.
- Activities define more specifically how the work will get done.
- A network diagram defines the sequence of how and when the activities will be performed.
- Project planning is a critical activity in developing an information system (IS).
- A project management planning tool or methodology, called the systems development life cycle (SDLC), is often used to help plan, execute, and control IS development projects.
- Numerous project management information systems are available to help project managers plan, track, and control projects in a completely interactive way.

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Summary

- The fourth and final phase of the project life cycle is closing the project.
- An important project closing action is staff recognition and evaluation.
- Conduct post project evaluation meetings within the project team or organisation with individual team members and a group meeting with the project team to gather team feedback for improved future performance.
- Identify and document lessons learnt to capitalise on the knowledge and experience gained on the project in order to improve performance on future projects.
- The project team or contractor should ensure that copies of appropriate project documentation are properly organised, filed, and archived so that they can be readily retrieved for future use.
- Conduct a post project evaluation meeting, or distribute a customer evaluation survey to the customer or sponsor to determine whether the project provided the customer with expected benefits; assess the level of customer satisfaction; and obtain any feedback that would be helpful in future business relationships with this customer or with other customers.
- Projects may be terminated before completion for various reasons.

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